

Web Scripting Midterm Exam Reviewer & Practice Quiz

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PART I: Identification (10 pts)

1. Define what a scripting language is.
2. Differentiate between client-side and server-side scripting.
3. Give three examples of client-side scripting uses.
4. Who conceptualized JavaScript and in what year?
5. What is the difference between ECMAScript and JavaScript?
6. Identify one JavaScript engine and where it is used.
7. What is the purpose of 'defer' in a script tag?
8. Name three DOM APIs and their main purpose.
9. What does 'addEventListener()' do in JavaScript?
10. What is the difference between '==' and '===' operators?

PART II: Multiple Choice (20 pts)

1. Which of the following runs on the web browser? (a) PHP (b) Python (c) JavaScript (d) MySQL
2. Which JavaScript variable declaration is block-scoped? (a) var (b) let (c) const (d) both b and c
3. What method removes a child node from its parent? (a) deleteNode() (b) removeChild() (c) popNode() (d) clearNode()
4. PHP is primarily executed on the: (a) Browser (b) Client (c) Server (d) IDE
5. What property lists only element children? (a) childNodes (b) children (c) nodes (d) childElements
6. Which DOM method finds an element by CSS selector? (a) querySelector() (b) getByClass() (c) selectCSS() (d) findElement()
7. What is true about JavaScript types? (a) Strongly typed (b) Statically typed (c) Dynamically typed (d) None
8. Which keyword prevents reassignment? (a) let (b) var (c) const (d) none
9. Which event triggers when the page fully loads? (a) input (b) load (c) submit (d) ready
10. The DOM represents: (a) Data Object Map (b) Document Object Model (c) Dynamic Object Module (d) Document Oriented Memory
11. Which of the following is NOT a primitive type in JS? (a) number (b) boolean (c) array (d) string
12. What function creates a new HTML element? (a) makeElement() (b) document.createElement() (c) document.newElement() (d) element.create()
13. JavaScript functions are: (a) first-class objects (b) constants (c) variables only (d) fixed declarations
14. Which of the following is a server-side scripting language? (a) JavaScript (b) PHP (c) CSS (d) HTML
15. Which method attaches an event listener? (a) addEventListener() (b) attach() (c) onClick (d) listen()
16. What will '5' + 5 result in? (a) 10 (b) 55 (c) Error (d) NaN
17. What is the result of typeof []? (a) array (b) object (c) list (d) collection

18. What happens when using '==='? (a) Compares only values (b) Compares values and types (c) Compares addresses (d) Converts before comparing
19. The official JavaScript standard specification is: (a) JS16 (b) ECMA2025 (c) ECMAScript (d) NodeScript
20. Which tag executes JS directly in HTML? (a) <script> (b) <javascript> (c) <js> (d) <code>

PART III: Code Tracing (15 pts)

1. What is the output of: `let x=5; let y='5'; console.log(x==y); console.log(x===y);`
2. Predict the DOM output: `let div=document.createElement('div'); let p=document.createElement('p'); p.textContent='Hello'; div.appendChild(p); document.body.appendChild(div);`
3. What does this log? `const arr=[1,2,3]; arr.push(4); console.log(arr.length);`
4. What happens when 'defer' is used in a script tag?
5. Trace the output: `function greet(name){return `Hi, ${name}`;} console.log(greet('Jec'));`

PART IV: Short Answer (15 pts)

1. Explain the purpose of the DOM tree in web development.
2. Describe how JavaScript interacts with HTML using the DOM API.
3. Compare client-side validation vs. server-side validation.
4. What are web scripting engines, and name two examples.
5. How does the browser execute JS when using `async` and `defer`?

■ ANSWER KEY

PART I: Sample Answers 1. Interpreted language needing a host. 2. Client runs in browser; server runs on server. 3. Validation, animation, DOM updates. 4. Brendan Eich, 1995. 5. ECMAScript=standard; JS=implementation. 6. V8 (Chrome), SpiderMonkey (Firefox). 7. Defer delays script execution until DOM parsed. 8. DOM, BOM, Fetch. 9. Adds event handler. 10. '===' compares type & value.

PART II: 1-c, 2-d, 3-b, 4-c, 5-b, 6-a, 7-c, 8-c, 9-b, 10-b, 11-c, 12-b, 13-a, 14-b, 15-a, 16-b, 17-b, 18-b, 19-c, 20-a.

PART III: 1. true, false. 2. <div><p>Hello</p></div>. 3. 4. 4. Defer waits for DOM before running. 5. Hi, Jec.

PART IV: 1. DOM represents HTML as manipulable objects. 2. JS changes HTML elements via DOM methods. 3. Client validation=user-side, server=backend-side. 4. Engines interpret JS (V8, SpiderMonkey). 5. Async runs ASAP; defer waits for DOM ready.